# FUNDAMENTALS OF AI AND ML EVALUATED COURSE PROJECT

# TITLE : Smart Traffic Signal Control System Using Python

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REG NO. : 25BCE10064

BRANCH : CSE(CORE)

YEAR : 2025

# Introduction

Traffic congestion is a common problem in urban areas, especially at intersections where signals operate on fixed timers. These fixed systems do not consider real-time traffic conditions, leading to unnecessary waiting time and inefficient traffic flow.

This project aims to simulate a smart traffic signal system using Python that dynamically adjusts signal timing based on vehicle density and prioritizes emergency vehicles.

# Problem Statement

Traditional traffic signals operate on predefined time intervals regardless of traffic volume. This leads to:

- Longer waiting times on empty or low-traffic roads.
- Traffic congestion on busy roads.
- Delay in movement of emergency vehicles

There is a need for a system that can adapt to real-time traffic conditions and to improve efficiency.

# Objective

The objective of this project is to:

- Simulate a smart traffic signal system.
- Adjust signal timing based on the number of vehicles.
- Provide priority to emergency vehicles.
- Store and analyze traffic data.

# Methodology

The system works based on the following steps:

1. User inputs the number of vehicles on four roads (A, B, C, D)
2. The system checks if there is any emergency vehicle
3. If an emergency is present, that road is given priority
4. Otherwise, roads are sorted based on vehicle count
5. Signal timing is calculated dynamically using a formula
6. Signals are displayed using a countdown timer
7. Data is stored in a CSV file for each cycle
8. Peak traffic time is calculated using stored data

# Key Features

- Dynamic signal timing based on traffic density
- Emergency vehicle priority handling
- Real-time signal simulation with countdown
- Data storage using CSV file
- Peak traffic analysis feature
- Handling of edge cases like zero traffic

# Technologies Used

- Python
- CSV file handling
- Time module for simulation
- Basic logic and control structures

# Challenges Faced

- Designing a realistic timing formula
- Handling edge cases such as zero or very high vehicle count
- Managing continuous simulation without errors
- Ensuring data is stored correctly

# Results

The system successfully simulates a traffic signal that adapts based on input traffic conditions. It reduces unnecessary waiting time for empty roads and prioritizes emergency vehicles effectively. The stored data also allows analysis of peak traffic periods.

# Conclusion

This project demonstrates how simple programming logic can be used to simulate real-world systems. The smart traffic signal improves efficiency compared to traditional fixed-timer systems. It also provides insights into how data can be used for traffic analysis.

# Future Scope

1. Integration with real-time sensors
2. Graphical user interface (GUI)
3. AI-based traffic prediction
4. Integration with smart city systems

# THANK YOU